Joel Villatoro

CONTACT INFORMATION

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EDUCATION

Department of Mathematics, University of Illinois

Ph.D. in Mathematics, August 2018

- Dissertation Title: Stacks in Poisson Geometry
- Advisor: Rui Fernandes

University of Oklahoma

B.S. in Mathematics, May 2012

ACADEMIC POSITIONS

NSF Postdoc, Washington University in St. Louis

Spring 2022-Present (Funded by MPS-Ascend grant, Award No. 2137999)

Guest, Max Planck Institute for Mathematics in Bonn, Germany

Fall 2021

Postdoctoral Fellow, KU Leuven in Belgium

September 2018-August 2021

HONORS AND AWARDS

2021	Awarded MPS Ascend Fellowship from the NSF
2017–2018	NSF AGEP (Alliances for Graduate Education and the Professoriate) Research support
2016	On UIUCs 'List of Teachers Ranked as Excellent'
2014	NSF AGEP Research support
2014	UIUC Research Board Award
2013	On UIUCs 'List of Teachers Ranked as Excellent'
2012–2013	UIUC GAAN (Graduate Assistance in Areas of National Need) Research Fellowship

PUBLICATIONS & PREPRINTS

Xiang Tang, Joel Villatoro, Representations up to homotopy as 2-sheaves, *In preparation.*

Angel Roman, Joel Villatoro, On the convolution algebras of double groupoids, *In preparation.*

Joel Villatoro, Holonomy integrations of Poisson manifolds of compact type, *In preparation.*

Joel Villatoro, On the integrability of Lie algebroids by diffeological spaces, arXiv:2309.07258 *(preprint).*

Joel Villatoro, On Sheaves of Lie-Rinehart structures, arXiv:2010.15463 *(preprint).*

Alfonso Garmendia, Joel Villatoro, Integration of Singular Foliations via Paths, Inter-

national Mathematics Research Notices, Volume 2022, Issue 23, December 2022, Pages 18401-18445.

J. Villatoro. Picard groups of b-symplectic manifolds. *Journal of Symplectic Geometry* (2021), vol. 19, no. 3, 723-775. https://dx.doi.org/10.4310/JSG.2021.v19.n3.a6

J. Villatoro, Stacks in Poisson Geometry. (2018 PhD Thesis). arXiv:1806.01939.

J. Villatoro. Poisson manifolds and their associated stacks. *Letters in Mathematical Physics* (2018), no. 3, 897-926. https://doi.org/10.1007/s11005-017-1012-5.

Teaching Experience

Washington University in St Louis

Fall	2022	**Differential equations (Instructor)** Undergraduate course in differential equations.
Spring	2022	**Calculus of Several Variables (Instructor)** Junior (300 level) course in multivariable calculus.

KU Leuven

Spring	2021	**Student Seminar Grader**: Evaluate and provide feedback on individual bachelor student presentations
Fall	2020	**Differential Equations(TA)**: Monitor and advise students as they work individually on assigned exercises.
Spring	2020	**Symplectic Geometry(Instructor)**: Graduate level course. Design a curriculum based on a syllabus of required topics. Give lectures, write and grade exams and homework. Switched to synchronous lectures after COVID.
Fall	2019	**Wiskunde 1(TA)**: Single variable calculus for non-mathematics majors. Monitor and advise students as they work individually on assigned exercises.

University of Illinois at Urbana-Champaign

Spring	2017	**Calculus I(TA)**: Single variable calculus. Prepare exercises and lead students in active learning sessions. Put students into groups and provide assistance and encourage cooperation and discussion.
Fall	2016	**Calculus I (TA)**: See above.
Fall	2013	**Calculus III (TA)**: Multivariable calculus. Same class format as Calculus I.
Fall	2012	**Linear Algebra (Tutor)**: Work in official university tutoring room for Linear Algebra students. Give hints and advise to students who come with questions and homework.

Supervision

Spring 2021	**Bachelor project supervisor**: Mentor a group of 6 Bachelor students in studying a new topic and preparing a presentation.
Spring 2018	**Illinois Geometry Lab**: graduate student assistant. Graduate student supervisor for a team of students working on computation based research project about Poisson structures on Euclidean space.
Fall 2017	**Illinois Geometry Lab**: graduate student assistant. Graduate student supervisor for a team of students working on computation based research project about modeling a partial differential equation.

Diversity Equity and Inclusion

Fall 2023	**Puertas College Prep**: Organizer and mentor for Junior and Senior level highschool students from disadvantaged backgrounds.
Spring 2023	**WashU Mathematics DEI Committee**: Served on the math department's committee.
Fall 2022	**WashU Mathematics DEI Committee**: Served on the math department's committee.
Fall 2022	**Puertas College Prep Mentor and Organizer**: Organizer and mentor for Junior and Senior level highschool students from disadvantaged backgrounds.
2015–2017	**SLOAN Peer Mentor**: Work with an assigned junior graduate student from underrepresented background. Provide mentorship, advise to mentee as they adapt to life as a new graduate student.
Summer 2012	**Project Transformation Camp Counselor**: Worked at an educational summer camp for low income children in the OKC metro area.

Organizational Experience

Fall 2023	UIUC-WUSTL Joint Seminar Organizer
Spring 2022	WashU Geometry Seminar Organizer
Fall 2022	WashU Geometry Seminar Organizer
Spring 2021	AGEP-GRS Conference Steering Committee
Fall 2020	Junior Seminar Organizer
Fall 2019-Spring 2020	KU Leuven, Weekly Poisson Meetings Organizer
Fall 2018-Spring 2019	KU Leuven, Weekly Poisson Meetings Organizer

Service

Spring 2023	**WashU Mathematics DEI Committee**: Served on the math department's committee.
Fall 2022	**WashU Mathematics DEI Committee**: Served on the math department's committee.
Spring 2022	**Washington University Math Club**: Organized meetings for the undergraduate mathematics club.
Spring 2022	**Glenride Elementary School Math Club**: Mathematics outreach and lessons at local elementary school.

Talks

Integration and differentiation for diffeological groupoids, 5-day workshop on Poisson Geometry, Lie Groupoids and Differentiable Stacks. BIRS, Banff (June 2022).
Recording: `https://www.birs.ca/events/2022/5-day-workshops/22w5035/videos/watch/202206101030-Villatoro.html`

Diffeological Solutions to the Integration Problem Gone Fishing (2020-2022). Savannah, GA (March 2022).

Diffeological vector bundles vs sheaves of modules, Western University. London, Ontario CA. (October 2020).

On the homotopy theory and algebraic geometry of sheaves of Lie-Rinehart algebras, Junior Global Poisson Workshop 2020. Online, (September 2020).
Recording: `https://www.youtube.com/watch?v=VFVr4xPkDRw&feature=emb_title`

A path approach to holonomy for singular foliations, University of Illinois and Urbana-Champaign. Urbana, Illinois USA. (December 2019).

Holonomy for Singular Foliations, University of Antwerp. Antwerp, Belgium. (November 2019).

Symplectic Morita Equivalence and Stacks, Max Planck Institute for Mathematics. Bonn, Germany (February 2019).

On the singularities of the Weinstein groupoid, Higher Geometric Structures along the Lower Rhine XII. Radboud University, Nijmegen, NL (January 2019).

Transversely Integrable Lie Algebroids and Asymmetric Orbispaces, Utrecht University. Utrecht, Netherlands (October 2018).

Geometric stacks for groupoids with geometry, Poisson 2018. University of Toronto, Toronto Canada (August 2018).

Picard groups of b-symplectic manifolds, 3rd Pacific Rim Mathematical Association (PRIMA) congress. Instituto Tecnológico de Oaxaca, Oaxaca Mexico (August 2017).

Morita type equivalences for non-integrable algebroids, Conference on Poisson Geometry and Stacks. The Fields Institute, University of Toronto (July 2017).

Poisson manifolds and their associated stacks, 5-day workshop on Geometric Structures on Lie Groupoids. BIRS, Banff (April 2017).
Recording: `https://www.youtube.com/watch?v=rEzlq8118CQ`

Poisson manifolds and their stacks, Symplectic and Poisson Geometry Seminar. University of Illinois at Urbana-Champaign (March 2017).

Picard Groups of b-Symplectic Manifolds Symplectic and Poisson Geometry Seminar. University of Illinois at Urbana-Champaign (October 2016).

Motivating stacks as models for singular spaces, Graduate Geometry and Topology seminar. University of Illinois at Urbana-Champaign (September 2016).

Poisson manifolds as families of Hamiltonian systems, Graduate Geometry and Topology seminar. University of Illinois at Urbana-Champaign (September 2015).